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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CSC 416 — Past Questions & Materials
by Renance

Course: CSC 416
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 400

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Paper 1 of 4 — CSC 416 FIRST SEMESTER 2023/2024

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF COMPUTER SCIENCE

1ST SEMESTER BSC EXAMINATION (2023/2024 SESSION)

CSC 416: COMPUTER COMPUTATIONAL SCIENCE AND NUMERICAL METHODS

INSTRUCTION: Attempt Questions 1, 2 and Any Other 2

TIME ALLOWED: 2½ Hrs.

Question 1 [30 Marks]

- a) Several definitions of operations research have been given by various experts and societies.
- Define at least two (2).
 - Describe the stages of operations research development.
- b) Most numerical analysts share some common concerns, perspectives, and mathematical methods of analysis.
- Describe any five (5) of these common perspectives.
 - State the characteristics of numerical computing.
 - How are numbers classified under numerical computation?
- c) Rounding errors are incurred by cutting-off large numbers into significant digits.
- Describe the five (5) types of rounding error.
 - Describe the concept of accuracy and precision in relation to significant digits.
 - Evaluate the sum $s = 13 + 15 + 17$ to four significant digits and find its absolute and relative error.

Question 2 (20 Marks)

- b) A farmer has 100 acres on which to plant two crops (corn or wheat). Production expenses is shown in the table 1. Each acre yields an average of 110 bushels of corn or 30 bushels of wheat. Available capital is \$15,000, and available storage facilities is 4,000 bushels. If net profit per bushel of corn and wheat is \$1.30 and \$2.00 respectively, how should the farmer plant the 100 acres to maximize the profits? Formulate the linear programming model for the problem below:

Question 3 [10 Marks]

- a) Briefly define the following terms:
i. Model. ii. Computer Model iii. Modeling and Simulation.
- b) List any seven (7) steps involved in developing a Model.

Cost Per Acre (#)	Item	
	Corn	Wheat
Seed	12	40
Fertilizer	58	80
Planting/Care/Harvesting	50	90
Total Expense	120	210

Question 4 [10 Marks]

- a) State the rules for Rounding off a number to the n^{th} significant digits. 295.10542
 b) Using the stated rules in 4(a):
 i. Round 295.10542 to 5 significant digits
 ii. Round 0.70029 to 4 significant digits
 iii. Calculate $\sqrt{102} - \sqrt{101}$ correct to four significant digits.

Question 5 [10 Marks]

- a) How are numbers classified under numerical computation? *exact approx*
b) Convert the following as indicated:
i. $45.12_{10} \rightarrow$ decimal ii. $15375_{10} \rightarrow$ binary iii. $12.88_{10} \rightarrow$ Octal

0.7 8 6 2 4

Paper 2 of 4 — CSC 416 PQ

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF COMPUTER SCIENCE
1ST SEMESTER BSC EXAMINATION (2023/2024 SESSION)
CSC 416: COMPUTER COMPUTATIONAL SCIENCE AND NUMERICAL METHODS

INSTRUCTION: Attempt Questions 1, 2 and Any Other 2

TIME ALLOWED: 2½ hrs

Question 1 [30 Marks]

- Several definitions of operations research have been given by various experts and societies.
 - Define at least two (2).
 - Describe the stages of operations research development.
- Most numerical analysts share some common concerns, perspectives, and mathematical methods of analysis.
 - Describe any five (5) of these common perspectives.
 - State the characteristics of numerical computing.
 - How are numbers classified under numerical computation?
- Rounding errors are incurred by cutting-off large numbers into significant digits.
 - Describe the five (5) types of rounding error.
 - Describe the concept of accuracy and precision in relation to significant digits.
 - Evaluate the sum $s = \sqrt{3} + \sqrt{5} + \sqrt{7}$ to four significant digits and find its absolute and relative error.

Question 2 [20 Marks]

- Define Linear programming (LP) as an operations research technique.
- A farmer has 100 acres on which to plant two crops (corn or wheat). Production expenses is shown in the table 1. Each acre yields an average of 110 bushels of corn or 30 bushels of wheat. Available capital is ₦15,000, and available storage facilities is 4,000 bushels. If net profit per bushel of corn and wheat is ₦1.30 and ₦2.00 respectively, how should the farmer plant the 100 acres to maximize the profits? Formulate the linear programming model for the problem below:

Table 1

Cost Per Acre (₦)	Item	
	Corn	Wheat
Seed	12	40
Fertilizer	38	80
Planting Care Harvesting	50	90
Total Expense	120	210

Question 3 [10 Marks]

- Briefly define the following terms:
 - Model.
 - Computer Model
 - Modeling and Simulation.
- List any seven (7) steps involved in developing a Model.

Question 4 [10 Marks]

- State the rules for Rounding off a number to the n^{th} significant digits.
- Using the stated rules in 4(a):
 - Round 295.10542 to 5 significant digits
 - Round 0.70029 to 4 significant digits
 - Calculate $\sqrt{102} - \sqrt{101}$ correct to four significant digits.

Question 5 [10 Marks]

- How are numbers classified under numerical computation?
- Convert the following as indicated:
 - $45.12_{10} \rightarrow \text{decimal}$
 - $15.375_{10} \rightarrow \text{binary}$
 - $12.88_{10} \rightarrow \text{Octal}$



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF COMPUTER SCIENCE
1ST SEMESTER BSC EXAMINATION (2023/2024 SESSION)
CSC 416: COMPUTER COMPUTATIONAL SCIENCE AND NUMERICAL METHODS

INSTRUCTION: Attempt Questions 1, 2 and Any Other 2

TIME ALLOWED: 2 1/2 Hrs

✓ Question 1 [30 Marks]

- a) Several definitions of operations research have been given by various experts and societies.
- Define at least two (2).
 - Describe the stages of operations research development.
- b) Most numerical analysts share some common concerns, perspectives, and mathematical methods of analysis.
- Describe any five (5) of these common perspectives.
 - State the characteristics of numerical computing.
 - How are numbers classified under numerical computation?
- c) Rounding errors are incurred by cutting-off large numbers into significant digits.
- Describe the five (5) types of rounding error.
 - Describe the concept of accuracy and precision in relation to significant digits.
 - Evaluate the sum $s = \sqrt{3} + \sqrt{5} + \sqrt{7}$ to four significant digits and find its absolute and relative error.

✓ Question 2 [20 Marks]

- a) Define Linear programming (L.P) as an operations research technique.
- b) A farmer has 100 acres on which to plant two crops (corn or wheat). Production expenses is shown in the table 1. Each acre yields an average of 110 bushels of corn or 30 bushels of wheat. Available capital is \$15,000, and available storage facilities is 4,000 bushels. If net profit per bushel of corn and wheat is \$1.30 and \$2.00 respectively, how should the farmer plant the 100 acres to maximize the profits? Formulate the linear programming model for the problem below:

Table 1

Cost Per Acre (\$)	Item	
	Corn	Wheat
Seed	12	40
Fertilizer	58	80
Planting/Care/Harvesting	50	90
Total Expense	120	210

✓ Question 3 [10 Marks]

- a) Briefly define the following terms:
- Model.
 - Computer Model
 - Modeling and Simulation.
- b) List any seven (7) steps involved in developing a Model.

✓ Question 4 [10 Marks]

- a) State the rules for Rounding off a number to the n^{th} significant digits.
- b) Using the stated rules in 4(a):
- Round 295.10542 to 5 significant digits
 - Round 0.70029 to 4 significant digits
 - Calculate $\sqrt{102} - \sqrt{101}$ correct to four significant digits.

✓ Question 5 [10 Marks]

- a) How are numbers classified under numerical computation?
- b) Convert the following as indicated:
- $45.12_8 \rightarrow$ decimal
 - $15.375_{10} \rightarrow$ binary
 - $12.88_{10} \rightarrow$ Octal

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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0.049629

Paper 3 of 4 — CSC 416 Computer Computational Science And Numerical Methods

Level 400 • First Semester • Handout

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A

CSC 416

Computer Computational Science and Numerical Methods

21/03/2025

Multiplication of Error Analysis

Show that $|x_1 x_2 - x_1 x_2| \leq |e_1| |x_2| + |e_2| |x_1|$ given that x_1 and x_2 are the value while x_1 and x_2 are approximate values and e_1 and e_2 are the errors involved

Solution

error = True value - approximate value

$e_1 = x_1 - x_1$ (i)

$e_2 = x_2 - x_2$ (ii)

from eqn ii make x_1 a sub of a formula

$x_1 = e_1 + x_1$ (iii)

from eqn iii make x_2 a sub of a formula

$x_2 = e_2 + x_2$ (iv)

multiply eqn iii by eqn iv

$x_1 = e_1 + x_1$

$x_2 = e_2 + x_2$

$(e_1 + x_1)(e_2 + x_2) = x_1 x_2$

$e_1 e_2 + e_1 x_2 + e_2 x_1 + x_1 x_2 = x_1 x_2$

Note: error should not be greater than one

$$X_1 X_2 - x_1 x_2 = e_1 x_2 + e_2 x_1 + e_1 e_2$$

Note: $e_1 e_2$ is negligible meaning its equal to 0

$$X_1 X_2 - x_1 x_2 = e_1 x_2 + e_2 x_1$$

Take the modulus

$$|X_1 X_2 - x_1 x_2| \leq |e_1| |x_2| + |e_2| |x_1|$$

Interpolation (Difference Table)

Gregory Newton method to evaluate Polynomial (the forward difference)

Find the polynomial $f(x)$, has the value

x	0	1	2	3	4	5	6
$f(x)$	-6	-2	8	36	96	194	348

Generate the difference Table, calculate the value of 3.6 using the difference table

$58x + 30x + 12x - 2x(-6) = 4$
 $-2x(-6) = 4$

X	f(x)	$\Delta^1 f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
0	-6	4	6	12	0
1	-2	10	18	12	0
2	8	28	30	12	0
3	36	58	42	12	0
4	94	100	54	12	0
5	194	154	66	12	0
6	364				

$\Delta = -2 - (-6)$
 $= -2 + 6 = 4$

when $\Delta^4 f(x)$ is 0 then the polynomial is x^3

Newton's formula

$$f(x) = f(a) + u \Delta f(a) + \frac{u(u-1)}{2!} \Delta^2 f(a) + \frac{u(u-1)(u-2)}{3!} \Delta^3 f(a) + \frac{u(u-1)(u-2)(u-3)}{4!} \Delta^4 f(x)$$

$x = 3.6$, $a = 0$ and $h = 1$
 $u = \frac{x-a}{h} = \frac{3.6-0}{1} = 3.6$

don't use group method
substituting

$$f(x) = 0 + \frac{3 \cdot 6 \Delta f(1)}{2} + \frac{3 \cdot (3 \cdot 6 - 1) \Delta^2 f(1)}{2}$$

$$= \frac{3 \cdot 6 (3 \cdot 6 - 1) (3 \cdot 6 - 2) \Delta^3 f(1)}{6} + \frac{3 \cdot 6 \times 2 \cdot 6 \times 6}{6} + \frac{3 \cdot 6 \times 2 \cdot 6 \times 6 \times 6}{6}$$

$$= -6 + \frac{3 \cdot 6 (4)}{2} +$$

$$= -6 + 14 \cdot 4 + 28 \cdot 08 + 29 \cdot 95$$

$$= 66.43$$

Numerical method of Quadratic Equation

$$ax^2 + bx + c$$

$$x_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{c}{ax_1}$$

$$(x^2 - 4 = 0)^2$$

$$x^2 - 4 = 0$$

if it equal to 0 it means it will
iteration converging & diverging

I used fixed point method (Repeated Procedure method) to evaluate
a root of Equation

Find the positive root of the equation $x^2 - 4x + 2 = 0$ at point

$x_0 = 3.2$ by fixed point (Repeated Procedure method)

Solution

$x^2 - 4x + 2 = 0$

$x^2 = 4x - 2$

$x = \sqrt{4x - 2}$

$x_{n+1} = \sqrt{4x_n - 2}$

when $n = 0$

$x_{n+1} = \sqrt{4x_0 - 2}$

$x_1 = \sqrt{(4 \times 3.2) - 2}^{1/2}$

$x_1 = \sqrt{(12.8 - 2)}^{1/2}$

$x_1 = (10.8)^{1/2}$

$x_1 = 3.2863353545$

when $n = 1$

$x_2 = \sqrt{4x_1 - 2}$

$x_2 = \sqrt{4(3.2863353545) - 2}$

(Define iteration
state the condition of
iteration method when it
is converge or diverge)

Newton Raphson method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{f'(x_n)}{f(x_n)}$$

Using Newton Raphson method find a root of a equation $x^2 - 4 = 0$, $x_0 = 1$

$$x_0 = 1$$

$$f(x) = x^2 - 4$$

$$f(x_0) = x_0^2 - 4$$

$$f'(x_0) = 2x_0$$

when $n = 0$

$$f(x_0) = x_0^2 - 4$$

$$f'(x_0) = 2x_0$$

when $x_0 = 1$

$$f(1) = 1^2 - 4$$

$$= 1 - 4 = -3$$

$$f'(1) = 2(1) = 2$$

Trapezoidal Rule wrote program for 50 years

The formula for trapezoidal rule is used to find the Area under the curve

Formula: Trapezoidal's rule = $\frac{h}{2} (\text{sum of first ordinate} + \text{last ordinate}) + 2(\text{sum of other ordinates})$

h is interval

$h = \frac{\text{upper bound} - \text{lower bound}}{\text{number of ordinates} - 1}$

= upper bound - lower bound

unequal space intervals $\rightarrow$ no will specify number

$P = P_0 e^{kt}$

e exponential

In the census of 2006 the population of delta is

$t = 20 \text{ years}$ $k = 3.8\%$ $k = \frac{3.8}{100} = 0.038$

t years

$P = 4112445 e^{0.038t}$

$P = \int_0^{120} 4112, 445 e^{0.038t}$

Note: evaluate

(i) Using integration method (ii) Using trapezoidal rule

$$= \frac{1}{0.038} \left[4112,445 e^{0.038(20)} - e^{0.038(0)} \right]$$

$$= \frac{1}{0.038} \left[4112,445 e^{0.76} - 1 \right]$$

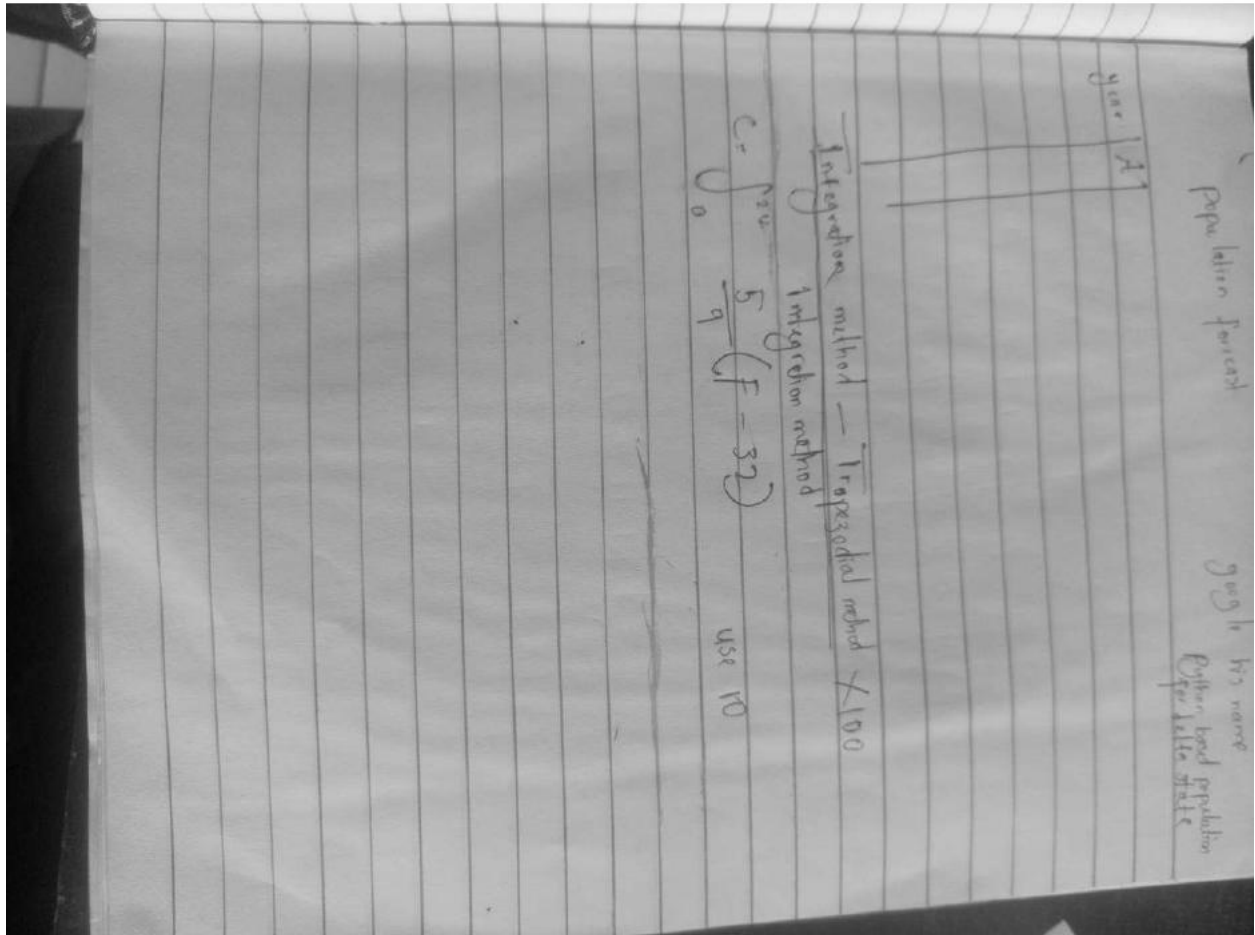
$$= 123,186,918.73$$

1 trapezoidal Rule

$$h = \frac{20-0}{10} = 2$$

t	$P = \int_0^{20} 4112,445 e^{0.038t}$
0	4112,445
20	4112,174,244

$y_0, y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9, y_{10}$



Paper 4 of 4 — CSC 416 handout

Level 400 • First Semester • Handout

Lecture notes on

COMPUTER COMPUTATIONAL SCIENCE AND
NUMERICAL METHODS

Csc 416

FIRST SEMESTER (2023/2024 SESSION)

DEPARTMENT OF COMPUTER SCIENCE,
FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

Operations
research

In our daily lives we make several decisions without even noticing them. These decisions are taken simply by common sense, judgment and expertise, without using any mathematical or any other model in simple situations. But the decisions we are concerned with here are complex and are heavily responsible, thus requiring some sort of substantiated model before decisions can be taken. Examples are public transportation network planning in a city having its own layout of factories and residential blocks, or finding the appropriate product mix when there exists a large number of products with different profit contributions and production requirements, etc.

Operations Research (O. R.) involves the application of mathematical and quantitative methods for decision making. The main purpose of Operations Research is to provide a rational basis for decision making in the absence of complete data, where systems composed of humans, machines, and procedures may not have the necessary information required to make complex decisions that impact everyday life. Operations Research is an interdisciplinary study which combines tools from different discipline such as mathematics, statistics, economics, psychology, engineering etc., to make a new set of knowledge for decision making.

Definition: Several definitions stressed by various experts and Societies on the subject matter enable us to know what Operations Research is, and what it does. They are as follows:

1. According to the Operational Research Society of Great Britain, Operational Research is the attack of modern science on complex problems arising in the direction and management of large systems of men, machines, materials and money in industry, business, government and defense. Its distinctive approach is to develop a scientific model of the system, incorporating measurements of factors such as change and risk, with which to predict and compare the outcomes of alternative decisions, strategies or controls. The purpose is to help management determine its policy and actions scientifically.
2. Randy Robinson stresses that Operations Research is the application of scientific methods to improve the effectiveness of operations, decisions and management. By means such as analysing data, creating mathematical models and proposing innovative approaches, Operations Research professionals develop scientifically based information that gives insight and guides decision making. They also develop related software, systems, services

and products.

3. Morse and Kimball have stressed Operations Research is a quantitative approach and described it as “ a scientific method of providing executive departments with a quantitative basis for decisions regarding the operations under their control”.

4. Saaty considers Operations Research as tool of improving quality of answers. He says, “O.R. is the art of giving bad answers to problems which otherwise have worse answers”.

5. Miller and Starr state, “O.R. is applied decision theory, which uses any scientific, mathematical or logical means to attempt to cope with the problems that confront the executive, when he tries to achieve a thorough-going rationality in dealing with his decision problem”.

6. Pocock stresses that O.R. is an applied Science. He states “O.R. is scientific methodology (analytical, mathematical, and quantitative) which by assessing the overall implication of various alternative courses of action in a management system provides an improved basis for management decisions”.

STAGES OF DEVELOPMENT OF OPERATIONS RESEARCH

Step I: Observe The Problem Environment

The first step in the process of O.R. development is the problem environment observation. This step includes different activities; conferences, site visit, research, observations etc., which provides sufficient information to the O.R. specialists on how to formulate the problem.

Step II: Analyse And Define The Problem

In addition to the problem definition, the objectives, uses and limitations of the O.R. study of the problem is also defined. The outputs of this step involve understanding the nature of the problem and outlining the need for a solution to the problem.

Step III: Develop A Model

A model is a representation of some abstract or real situation. The models are basically mathematical models, which describes systems, processes in the form of equations, formula/relationships. The different activities in this step are variables definition, formulating equations, etc. The model is then tested in the field under different environmental constraints and modified accordingly.

Step IV: Select Appropriate Data Input

A model works appropriately when there is appropriate data input. The activities in this step include internal/external data analysis, fact analysis, and collection of opinions and use of computer data banks. The objective of this step is to provide sufficient data input to operate and test the model developed in Step III.

Step V: Provide A Solution And Test Its Reasonableness

This step involves providing a solution with the help of the model and input data. This solution is not implemented immediately, instead it is used to test the model and identify limitations (if any). If the solution is not reasonable or the behaviour of the model is not proper, the model is updated and modified at this stage.

Step VI: Implement The Solution

At this step the solution obtained from Step V is implemented.

OPERATIONS RESEARCH TECHNIQUES – LINEAR PROGRAMMING

Many management decisions involve trying to make the most effective use of organization resources. These resources include Machinery, Labours, Money, Time, Warehouse space or Raw materials to produce goods (machinery, furniture, food or cooking) or service (schedules for machinery and production advertising policies or investment decision). Linear Programming (LP) is a widely used mathematical techniques designed to help managers in planning and decision making relative to resource allocations.

Linear programming (LP) or Linear Optimisation may be defined as the problem of maximizing or minimizing a linear function that is subjected to linear constraints. The constraints may be equalities or inequalities. The optimisation problems involve the calculation of profit and loss. Linear programming problems are an important class of optimisation problems, that helps to find the feasible region and optimise the solution in order to have the highest or lowest value of the function.

All Linear Programming Models Have Four Basic Properties In Common.

1. All LP models seek to maximize or minimize some quantity, usually profit or costs.
2. All LP models have constraints or limitations that limit the degree to which the objective can be achieved.
3. There must be alternative course of action to choose from e.g. if there are 4 different product, management may decide (using LP) how to allocate limited resources among them.
4. Objectives and constraints in LP model must be expressed in linear equations and inequalities.

Application Of Linear Programming To Business

- Product-Mix – LP is used in the selection of the product-mix in a factory to make best use of machine and machine hours available while maximizing profit, that is, to find out which product to include in production plan and in what quantities that should be produced.
- Blending Problems – LP is used for the selection of different blends of raw materials to produce the best combinations at minimum cost e.g. food drinks, etc.
- Production Scheduled – LP is used to develop a production scheduled that will satisfy future demands for a firm's product and at the same time minimize production and inventory cost.
- Production Quantity – LP is used in determining how much quantity of different grades of product to produce in order to yield maximum profit.
- Distribution System – LP is used in determining a distribution system that will minimize total shipping cost from several warehouses to various market locations.
- Limited Advertisement – LP is used in the allocation of limited advertising budget among radio, TV and newspaper spots in order to maximize the returns on investment.

- Investment – LP is used in selecting investment port-folio from a variety of stocks and bonds
- available in such a way as to maximize the returns on investment.
- Work Scheduled – LP is used in the development of a work scheduled that allows a large restaurant to meet staff needs at all hours of the day, while minimizing the total number of employees

Assumption of Linear Programming

- Certainty – We assume that numbers in the objective and constraints are known with certainty and does not change during the period being study.
- Proportionality – We are sure that proportionality exists in the objective and the constraints. This means that, if production of one unit of product uses two of a particular scare resource; then making five units of that product uses ten resources.
- Additivity – This means that the total of all activities equals the sum of each individual activity.
- Divisibility – This means that solution may take fractional values and need not be in whole numbers (integers). If a fraction of a product cannot be produced, integer programming problem exist.
- Non-negativity – We assume that all answers or variables are non-negative. Negative values of physical quantities are an impossible solution.

Formulation of Linear Programming Model

Linear programming problems are optimization problems which are stated verbally or in words. The following steps can be used to formulate the model of any optimization problem.

Step 1 – Choose variables and notations that will be used to form the objective and constraints functions.

Step 2 – Identify the objective function to either maximize or minimize e.g., maximizes profit or minimizes cost.

Step 3 – Develop mathematical relationships to describe objective and constraints.

Examples on Formulation Linear Programming Models

Example 1:

A farmer has 100 acres on which to plant two crops: corn or wheat. To produce these crops, there are certain expenses as shown in the table below. After the harvest, the farmer must store the crops awaiting proper market conditions. Each acre yields an average of 110 bushels of corn or 30 bushels of wheat. The limitations of resources are as follows: available capital is \$15,000, and available storage facilities is 4,000 bushels. If net profit (the profit after all expenses have been subtracted) per bushel of corn is \$1.30 and for wheat is \$2.00, how should the farmer plant the 100 acres to maximize the profits? Formulate the linear programming model for the above problem.

Item

Cost Per Acre (\$) Corn Wheat

Seed 12 40

Fertilizer 58 80

Planting/Care/Harvesting 50 90

Total Expense 120 210

Solution:

Formulate a mathematical model.

let x = Number of acres to be planted in corn.

let y = Number of acres to be planted in wheat.

The number of acres to be planted cannot be negative. Therefore:

$$x \geq 0 \text{(1)}$$

$$y \geq 0 \text{(2)}$$

The amount of available land is 100 acres. Therefore:

$$x + y \leq 100 \text{(3)}$$

The total expense for planting corn and wheat is 120 and 210 respectively. Therefore:

$$\text{Expenses for planting the corn} = 120x$$

$$\text{Expenses for planting the wheat} = 210y$$

The total expenses cannot exceed \$15,000. Therefore, the available capital is:

$$120x + 210y \leq 15000 \text{(4)}$$

Each acre yields an average of 110 and 30 bushels of corn and wheat respectively. Therefore:

$$\text{Yield of acreage planted in corn} = 110x$$

$$\text{Yield of acreage planted wheat} = 30y$$

The total yield cannot exceed the storage capacity of 4,000 bushels. Therefore:

$$110x + 30y \leq 4000 \text{(5)}$$

The farmer wants to maximize profit:

$$\text{Profit from corn} = \text{Profit per acre} \times \text{Number of acres} = 1.30 \times 110x = 143x$$

$$\text{Profit from wheat} = \text{Profit per acre} \times \text{Number of acres} = 2.00 \times 30y = 60y$$

Let P = Total Profit. Therefore:

$$P = \text{Profit from corn} + \text{profit from wheat}$$

$$P = 143x + 60y \text{(6)}$$

Therefore, the linear programming model is stated as follows:

$$\text{Maximize: } P = 143x + 60y$$

$$\text{Subject to: } x + y \leq 100 \text{ [Available land]}$$

$$120x + 210y \leq 15000 \text{ [Available capital]}$$

$$110x + 30y \leq 4000 \text{ [Storage capacity]}$$

$$x \geq 0 \text{ [Non-negativity]}$$

$$y \geq 0 \text{ [Non-negativity]}$$

Formulate the LP model for the following problems

Remember to identify the objective function to either maximize profit or minimize cost.

1. KEVWE & SON LTD manufactures two types of settees; half-upholstered and full-upholstered. The contribution per unit to profit is \$80 for half-upholstered and \$90 for full-

upholstered. The number of materials needed per product and maximum available materials are given below.

Item

Product Wood Form Cover

Half-upholstered 2 2 5

Full-upholstered 1 4 5

Maximum available 12 24 35

2. A convalescent hospital wishes to provide at a minimum cost, a diet that has a minimum of 200g of carbohydrates, 100g of protein and 120g of fats per day. These requirements can be met with two foods:

Food Carbohydrates Protein Fats

A 10g 2g 3g

B 5g 5g 4g

If food A cost 29k per ounce and food B cost 15k per ounce, how many ounces of each food should be purchased for each patient per day in order to meet the minimum requirements at the lowest cost?

3. ZICCO Inc. is an investment company doing an analysis of the pension fund for a certain company. A maximum of \$10 million is available to invest in two places. No more than \$8 million can be invested in stocks yielding 12% and at least \$2 million can be invested in long-term bonds yielding 8% . The stock-to-bond investment ratio cannot be more than 1 to 3 . How should ZICCO advise their client so that the pension fund will receive the maximum yearly return on investment?

4. Suppose an industry is manufacturing two types of products P1 and P2. The profits per Kg of the two products are \$30 and \$40 respectively. These two products require processing in three types of machines. The following table shows the available machine hours per day and the time required on each machine to produce one Kg of P1 and P2. Formulate the problem in the form of linear programming model.

Profit/Kg P1 (\$30) P2 (\$40) Total Available Machine
hours/day

Machine 1 3 2 600

Machine 2 3 5 800

Machine 3 5 6 1100

5. A company owns two flour mills viz. A and B, which have different production capacities for high, medium and low-quality flour. The company has entered a contract to supply flour to a firm every month with at least 8, 12 and 24 quintals of high, medium and low quality respectively. It costs the company \$2000 and \$500 per day to run mill A and B respectively. On a day, Mill A produces 6, 2 and 4 quintals of high, medium and low-quality flour, Mill B produces 2, 4 and 12 quintals of high, medium and low-quality flour respectively. How many days per month should each mill be operated in order to meet the contract order most economically.

A brief explanation of other frequently used Operation Research tools/techniques are as follows:

Game Theory:

This is used for making decisions under conflicting situations where there are one or more players/opponents. In this the motive of the players are dichotomized. The success of one player tends to be at the cost of other players and hence they are in conflict.

Decision Theory:

Decision theory is concerned with making decisions under conditions of complete certainty about future outcomes (making some probability about what will happen in future).

Queuing Theory:

This is used in situations where a queue is formed. For example; customers awaiting a service, aircrafts waiting for landing, jobs waiting for processing in a computing device, etc. The objective here is to minimize the cost of waiting without increasing the cost of service.

Inventory Models:

This is used to make decisions that minimize total inventory cost. This model successfully reduces the total cost of purchasing, carrying, and out of stock inventory.

Simulation:

This is a procedure that studies a problem by creating a model of the process involved in the problem and then through a series of organized trials and error solutions attempt to determine the best solution. This can sometimes be a difficult/time consuming procedure. Simulation is used when actual experimentation is not feasible or solution of model is not possible.

Non-linear Programming:

This is used when the objective function and the constraints are not linear in nature. Linear relationships may be applied to approximate non-linear constraints but limited to some range, because approximation becomes poorer as the range is extended. Thus, the non-linear programming is used to determine the approximation in which a solution lies and then the solution is obtained using linear methods.

Dynamic Programming:

Dynamic programming is a method of analysing multistage decision processes whereby each elementary decision depends on those preceding decisions and as well as external factors.

Markov Process:

This is a random process indexed by time, and with the property that the future is independent of the past, given the present. Markov process is a stochastic process with the following properties:

- The number of possible outcomes or states is finite.
- The outcome at any stage depends only on the outcome of the previous stage.
- The probabilities are constant over time

Network Scheduling:

This technique is used extensively to plan, schedule, and monitor large projects (for example computer system installation, R & D design, construction, maintenance, etc.). The aim of this technique is minimizing trouble spots (such as delays, interruption, production bottlenecks, etc.) by identifying the critical factors. The different activities and their relationships of the entire project are represented diagrammatically with the help of networks and arrows, which is used for

identifying critical activities and path. There are two main types of technique in network scheduling, they are:

- Program Evaluation and Review Technique (PERT) – is used when activity time is not known accurately. Only probabilistic estimate of time is available.
- Critical Path Method (CPM) – is used when activity time is known accurately.

Information Theory:

The objective of this theory is to evaluate the effectiveness of flow of information with a given system. This is used mainly in communication networks but also has indirect influence in simulating the examination of business organizational structure with a view of enhancing flow of information.

ADVANTAGES OF OPERATIONS RESEARCH

1. Ensures Better Control – Organisations can maintain better control over their employees due to operations research. Operations management research establishes standards based on performance and measures productivity. Hence, to identify problem areas, business managers can track deviations from the standard. This ensures no time is wasted and corrective action can be taken immediately.
2. Provides Detailed Analysis – One of the reasons for the significance of operations management is that it relies on analytics. Mathematical and scientific methods are used to analyse and tackle various problems. Using these methods, operations research provides a detailed and insightful analysis which allows businesses to solve problems thoroughly with comprehensive solutions. This also helps them know how they can analyse similar problems if they happen again in the future.
3. Better Decision Making – The mathematical approach of operations management research allows people to analyse many more alternatives and constraints than the traditional intuitive approach. Hence, operations research makes it easier for businesses to analyze multiple alternatives easily and quickly. This ultimately results in deciding the optimal choice more confidently.
4. Minimal Uncertainties – As we mentioned before, operations research uses proven methods and modelling techniques. This helps companies to eliminate any uncertainties that may arise. When we input realistic data into an already effective problem-solving model, it can significantly reduce and even eliminate uncertainties for companies. With reliable data, resolving challenges and managing complex business processes becomes easier.
5. Better Co-ordination of Departments – The importance of operations research extends to smooth working between different departments too. With operation research analysis, departments like marketing and production can work together to increase overall productivity.
6. Boosts Business Productivity – Another important advantage of operation research that attracts various businesses is increased productivity. The mathematical formulas used in operations management research offer a variety of optimal choices of factory size, inventory mix, manpower planning, implementing new technologies and so on. This ensures that work is done faster.

LIMITATIONS OF OPERATIONS RESEARCH

1. Magnitude of Calculations – The aim of the O.R. is to find out optimal solution taking into consideration all the factors. In this modern world these factors are enormous and expressing them in quantitative model and establishing relationships among these require voluminous calculations, which can be handled only by machines.
2. Money and Time Costs – The basic data are subjected to frequent changes and incorporating these changes into the operations research models are very expensive. However, a fairly good solution at present may be more desirable than a perfect operations research solution available in future or after some time.
3. Non-quantifiable Factors – When all the factors related to a problem can be quantifiable only then operations research provides solution otherwise not. The non-quantifiable factors are not incorporated in O.R. models. Importantly O.R. models do not take into account emotional factors or qualitative factors.
4. Implementation – Once the decision has been taken it should be implemented. The implementation of decisions is a delicate task. This task must take into account the complexities of human relations and behaviour, and sometimes only the psychological factors.

Numerical
computation

INTRODUCTION

Numerical computation or numerical computing is an approach for solving complex mathematical problems using only simple arithmetic operations. The approach involves formulating mathematical models for physical situations that can be solved with arithmetic operations. It requires development, analysis and use of algorithms.

Numerical computations invariably involve a large number of arithmetic calculations and, therefore, require fast and efficient computing devices. The microelectronic revolution and the subsequent development of high, low-cost personal computers have had a profound impact on the application of numerical computing methods to solve scientific problems.

COMMON PERSPECTIVES IN NUMERICAL COMPUTING

Numerical computation is concerned with all aspects of the numerical solution of a problem, from the theoretical development and understanding of numerical methods to their practical implementation as reliable and efficient computer programs. Most numerical analysts specialize in small subfields, but they share some common concerns, perspectives, and mathematical methods of analysis. These include the following:

1. When presented with a problem that cannot be solved directly, they try to replace it with a “nearby problem” that can be solved more easily. Examples are the use of interpolation in developing numerical integration methods and root-finding methods.
2. There is a widespread use of the language and results of linear algebra, real analysis, and functional analysis (with its simplifying notation of norms, vector spaces, and operators).
3. There is a fundamental concern with error, its size, and its analytic form. When approximating a problem, it is prudent to understand the nature of the error in the computed

solution. Moreover, understanding the form of the error allows creation of extrapolation processes to improve the convergence behaviour of the numerical method.

4. Numerical analysts are concerned with stability, a concept referring to the sensitivity of the solution of a problem to small changes in the data or the parameters of the problem.

Consider the following example:

The polynomial $p(x) = (x - 1)(x - 2)(x - 3)(x - 4)(x - 5)(x - 6)(x - 7)$, or expanded, $p(x) = x^7 - 28x^6 + 322x^5 - 1,960x^4 - 6,769x^3 - 13,132x^2 + 13,068x - 5,040$ has roots that are very sensitive to small changes in the coefficients. If the coefficient of x^6 is changed to -28.002 , then the original roots 5 and 6 are perturbed to the complex numbers $5.459 + 0.540i$ —a very significant change in values. Such a polynomial $p(x)$ is called unstable or ill-conditioned with respect to the root-finding problem. Numerical methods for solving problems should be no more sensitive to changes in the data than the original problem to be solved. Moreover, the formulation of the original problem should be stable or well-conditioned.

5. Numerical analysts are very interested in the effects of using finite precision computer arithmetic. This is especially important in numerical linear algebra, as large problems contain many rounding errors.

6. Numerical analysts are generally interested in measuring the efficiency (or “cost”) of an algorithm. For example, the use of Gaussian elimination to solve a linear system $Ax = b$ containing n equations will require approximately $2n^3/3$ arithmetic operations. Numerical analysts would want to know how this method compares with other methods for solving the problem.

CHARACTERISTICS OF NUMERICAL COMPUTING

Accuracy – Every method of numerical computing introduces errors. They may be either due to using an appropriate in place of an exact mathematical procedure or due to inexact representation and manipulation of numbers in the computer. These errors affect the accuracy of the results.

Efficiency – One more consideration in choosing a numerical method for solution of a mathematical model is efficiency. It means the amount of effort required by both human and computer to implement the method.

Numerical Stability – Another problem introduced by some numerical computing method is that of numerical instability. Errors introduced into a computation, from whatever source, propagate in different ways. In some cases, these errors tend to grow exponentially, with disastrous computational results.

NUMERICAL METHODS

Numerical methods are algorithms that compute approximate solutions to a number of problems for which exact solutions are not available. These include, but are not limited to:

- evaluating functions;
- computing the derivative of a function;
- computing the integral of a function;
- solving a non-linear equation;

- solving a system of linear equations;
- finding the maximum or minimum of a function;
- finding eigenvalues of an operator;
- finding solutions to an ODE (ordinary differential equation);
- finding solutions to a PDE (partial differential equation).

Such algorithms can be implemented (i.e., programmed) on a computer.

Given a physical model, one first derives a mathematical model that describes the physical phenomenon, with suitable assumptions. This process is called mathematical modeling. The mathematical model is then analyzed and solved with suitable numerical methods, to obtain approximate solutions. The numerical results may be plotted and compared to possible physical experiments, to verify the mathematical model and to gain insight for improvement. The diagram below shows how various aspects are related.

NUMERICAL COMPUTING PROCESS

1. Formulation of a mathematical model.
2. Construction of an appropriate numerical method.
3. Implementation of the method to obtain a solution.
4. Validation of the solution.

NUMBERS & THEIR ACCURACY

For the sake of numerical computation, all numbers can be classified under two broad headings: exact numbers and approximate numbers.

Exact and Approximate Numbers

Exact numbers are fully represented by some digits. Examples include integers and rational numbers that can and have been completely written (e.g., 3.2158).

Approximate numbers are those that are not fully specified by the digits representing them. As an example, we could write the rational number $3/7$ as 2.3333. But the actual number is not exactly 2.3333.

Note:

- Rational numbers can be written as a fraction of two integers. Although certain rational numbers are exact numbers, a lot of rational numbers cannot be written as exact numbers as in the example above.
- Irrational numbers cannot be written as the ratio of any two integers (e.g., π).
- There are two families of numbers that are unending: the ones that repeat certain sequences (e.g., 12.345454545), and the ones that do not (e.g., 18.127849342.).

On that note, the order of preference in dealing with numbers in numerical computations is as follows:

1. Natural numbers,
2. Rational numbers that have a finite string of digits,
3. Rational numbers that have unending strings of digits, and
4. Irrational numbers.

Representation Of Numbers In Different Bases

Historically, there have been several bases for representing numbers, used by different civilizations. These include:

- 10: decimal, which has become standard nowadays;
- 2: binary, computer use;
- 8: octal;
- 16: hexadecimal, used in ancient China;
- 20: vigesimal, used in ancient France (numbers 70 to 79 are counted as 60+10 to 60+19 in French, and 80 is 4×20);
- 60: sexagesimal, used by the Babylonians.

In principle, one can use any integer (or even non-integer) β as the base. The value of a number in base β is then written as

Taking $\beta = 10$, we obtain the standard decimal representation.

The above formula allows us to convert a number in any base β into decimal base. In principle, one can convert the numbers between different bases. Here are some examples.

Example 1. Conversion from octal $\rightarrow$ decimal:

$$(45.12)_8 = (4 \times 8^1) + (5 \times 8^0) + (1 \times 8^{-1}) + (2 \times 8^{-2}) = (37.15625)_{10}.$$

Example 2. Conversion of octal $\leftrightarrow$ binary:

Observe that, since $8 = 2^3$, we have

- (1) $_8 = (001)_2$
- (2) $_8 = (010)_2$
- (3) $_8 = (011)_2$
- (4) $_8 = (100)_2$
- (5) $_8 = (101)_2$
- (6) $_8 = (110)_2$
- (7) $_8 = (111)_2$
- (10) $_8 = (1000)_2$.

Then, the conversion between these two bases become much simpler. To convert from octal to binary, we can convert each digit in the octal base, and write it into binary base, using 3 digits in binary for each digit in octal. For example:

To convert a binary number to an octal one, we can group the binary digit in groups of 3, and write out the octal base value for each group. For example:

Example 3. Conversion from decimal $(22.45)_{10} \rightarrow$ binary.

This example is of particular interest. Since the computer uses binary base, how would the number $(12.45)_{10}$ look like in binary base? How does a computer store this number in its memory?

The conversion takes two steps:

For the integer part, we keep dividing by 2, and store the remainders of each step, until we cannot divide anymore. Then, collect the remainder terms in the reversed order. Thus, we have remainder

2 22

2 11 0 □ (22)₁₀ = (10110)₂

2 5 1

2 2 1

2 1 0

0 1

For the fractional part, we will keep multiplying by 2, and store the integer part for the result.

Thus, we have:

remainder

0.45 × 2

.9 × 2 0

.8 × 2 1

.6 × 2 1

.2 × 2 1

.4 × 2 0 □ (0.45)₁₀ = (0.0111001100....)₂

.8 × 2 0

.6 × 2 1

.2 × 2 1

.4 × 2 0

.8 × 2 0

....

Putting them together, we get (22.45)₁₀ = (10110.0111001100....)₂

Significant Digits (Figures)

Digits that are used to express a number are called significant digits/figures. The following statements describe the notion of significant digits.

- All non-zero digits are significant.
- All zeroes occurring between non- zero digits are significant.
- Trailing zeroes following a decimal point are significant (e.g., 3.50, 65.0, 0.230 all have three significant digits).
- Zeroes between the decimal point which precedes non-zero digits are not significant (e.g., 0.01234, 0.001234, 0.0001234 all have four significant digits).
- When the decimal point is not written, trailing zeroes are not considered to be significant (e.g., 4500 contains only two significant digits).

Note the following rules:

1. The leftmost non-zero digit is the most significant digit (e.g., In 0.001243, 1 is the most significant digit).
2. If there is no decimal point, the rightmost non-zero digit is the least significant (e.g., in 145630000, 3 is the least significant figure).
3. If there is a decimal point, the rightmost digit is the least significant, even if it is zero (e.g., in 235.34200, the last 0 is the least significant).

4. All digits between the least significant and the most significant (inclusive) are significant (e.g., in both examples under rule 1 and 2, digits 1243 and 14563 are significant; 0.00004 has one significant figure; 984.13245 has 8 significant figures).

However, there is an exception: When a zero is obtained by rounding off (e.g., 329.5 is rounded to 3 significant figures and becomes 330), the last zero becomes significant in this case. You can compare this with rule 2 above.

Accuracy & Precision

The concept of accuracy and precision are closely related to significant digits.

□ Accuracy refers to the number of significant digits in a value (e.g., the number 57.345 is accurate to five significant digits).

□ Precision refers to the number of decimal positions, i.e., the order of magnitude of the last digit in a value (e.g., 57.396 have a precision of 0.001). The precision of a number is an indication of the number of digits that have been used to express it. In scientific computing, it is the number of significant digits or numbers, while in financial systems, it is the number of decimal places. Most currencies in the world are quoted to two decimal places.

Rounding Off

In numerical computation, we come across numbers which have large numbers of digits (irrational numbers are a perfect example of numbers with unending digits) and it will be necessary to reduce the number of digits by which a number is represented. This process is known as rounding off.

It might be necessary to round off our numbers to make them useful for numerical computation because it would require an infinite computer memory to store an unending number.

Rounding Errors

The error caused by cutting-off a large number into a usable figure is called rounding error or round-off error. Rounding errors are incurred by truncating a sequence of digits representing a number. Representing a rational number $7/3$ will result in an unending number (2.3333333.....) which will require us to round-off to the n th significant digit and might lead to round-off error. Apart from being unable to write this number in an exact form by hand, a calculator or computer can only handle a finite string of digits.

Rules for Rounding off a number to the n significant figures

To round off a number to n significant digits;

1. Discard all numbers to the right of the n th significant digit.
2. If the discarded number is
 - a) less than half a unit, leave the n th significant digit unchanged.
 - b) greater than half a unit, increase the n th significant digit by one.
 - c) equal to half a unit, leave the n th significant digit unchanged if it is even, otherwise increase by one.

When a number is rounded off in accordance with the rules stated above, then it is said to be correct to n significant figures.

Types of Errors

1. Inherent Error – As the name implies, these are errors that are inherent in the statement of the problem itself. This could be due to the limitations of the means of calculation (e.g., the calculator or the computer). This error could be reduced by using a higher precision of calculation.

2. Truncation Error – This is the error associated with truncating a sequence or by terminating an iterative process. For instance, this kind of error also results when we carry out numerical differentiation or integration, because we are replacing an infinite decimal process with a finite one.

3. Absolute Error – This is the numerical difference between the true value and approximate value of a quantity. Thus,

$$\text{Absolute Error (AE)} = \text{True Value (TV)} - \text{Approximate Value (AV)}$$

4. Relative Error – This is the ratio of the absolute error to a true value of that quantity being concerned. Thus,

$$\begin{aligned} \text{Relative Error (RE)} &= \frac{\text{Absolute Error (AE)}}{\text{True Value (TV)}} \\ \text{Relative Error (RE)} &= \frac{\text{Absolute Error (AE)}}{\text{True Value (TV)}} \end{aligned}$$

5. Percentage Error – This is the percentage value of relative error. Thus,

$$\begin{aligned} \text{Percentage Error (PE)} &= \text{Relative Error (RE)} \times 100\% \\ \text{Percentage Error (PE)} &= \frac{\text{Absolute Error (AE)}}{\text{True Value (TV)}} \times 100\% \end{aligned}$$

Examples

□ Round the following to the number of significant digits indicated.

o 3.142857143 to 5 significant figures

o Solution: The nth digit is even and the discarded number is equal to half a unit. Following rules 1 and 2c, the nth significant digit is unchanged. Therefore, 3.142857143 to 5 significant digits = 3.1428.

o 1.4123519 to 5 significant figures

Solution: The nth digit is odd and the discarded number is equal to half a unit. Following rules 1 and 2c, the nth significant digit is increased by one. Therefore, 3.142857143 to 5 significant digits = 1.4124.

o 295.10542 to 5 significant digits

o 0.0045829 to 3 significant digits

o 6.32431925 to 8 significant digits

o 12.0234831 to 1 significant digit

o 0.00022218 to 4 significant digits

o 0.70029 to 4 significant digits

o 2.36425 to 5 significant digits

□ Round off the following numbers to two decimal places: 2.3742, 81.255, 52.275, 48.21416.

□ Calculate $102 - 101$ correct to four significant digits.

□ A student measured the length of a string of actual length 72.5 cm as 72.4 cm. Calculate the

absolute error and the percentage error.

□ An approximate value of π is given as 3.1428571 and its true value is 3.1415926. Find EA and ER.

$2 + \sqrt{3}$

□ One of the exact roots of a certain quadratic equation is $\sqrt{2}$. For this root, find, correct to

1 significant figure, the percentage error that arises if the value used for $\sqrt{3}$ is rounded to 1 decimal place.

□ If $u = 3v^7 - 6v$, find the percentage error in u if $v=1$, and error in v is 0.05.

□ Conversion from octal $\rightarrow$ decimal:

o $(45.12)_8$,

o $(50.455)_8$,

o $(332.4)_8$,

o $(123344)_8$.

□ Conversion from decimal $\rightarrow$ binary:

o $(12.45)_{10}$

o $(15.375)_{10}$

o $(6.986)_{10}$

□ Conversion from decimal $\rightarrow$ Octal:

o $(473)_{10}$

o $(416)_{10}$

o $(350)_{10}$

Modeling

and

simulation

INTRODUCTION

Modern science would be inconceivable without computers to gather data and run model simulations. Whether it involves bringing back pictures of the surface of the planet Mars or detailed images to guide brain surgeons, computers have greatly extended our knowledge of the world around us and our ability to turn ideas into engineering reality. Thus, modelling and computer simulation are important interdisciplinary tools.

Modelling and Simulation is a discipline for developing a level of understanding of the interaction of the parts of a system, and of the system as a whole. The level of understanding which may be developed via this discipline is seldom achievable via any other discipline.

WHAT IS A MODEL?

□ A model in general is a pattern, plan, representation (especially in miniature), or description designed to show the main object or workings of an object, system, or concept. Stated otherwise, models are abstractions of real or imaginary worlds we create to understand their behaviour, play with them by performing "what if" experiments, make projections, animate or simply have fun.

□ A model (physical or hypothetical) is a representation of real-world phenomenon or elements (objects, concepts or events). Stated otherwise a model is an attempt to express a possible structure of physical causality.

□ Models in science are often theoretical constructs that represent any particular thing with a set of variables and a set of logical and or quantitative relationships between them. Models in this sense are constructed to enable reasoning within an idealized logical framework about these processes and are an important component of scientific theories.

□ A computer model is a simulation or model of a situation in the real world or an imaginary world which has parameters that the user can alter. For example: Newton considers movement (of planets and of masses) and writes equations that makes the dynamics intelligible, among which is $f = ma$ (where f = force, m = mass and a = acceleration). Newton by this expression makes a formidable proposition, that force causes acceleration, with mass as proportionality coefficient. Another example is a model of an airplane which is a physical representation of the real airplane. Model airplanes are useful in predicting the behaviour of real airplanes when subjected to different conditions; weather, speed, load, etc. Computer models help us frame our thinking about objects in the real world. It should be noted that more often than not we model dynamic (changing) systems.

Why to use models?

To fly a simulator is safer and cheaper than the real airplane. For precisely this reason, models are used in industry commerce and military: it is very costly, dangerous and often impossible to make experiments with real systems. Provided that models are adequate descriptions of reality (they are valid), experimenting with them can save money, suffering and even time.

WHAT IS MODELING?

□ Modeling is the process of producing a model; a model is a representation of the construction and working of some system of interest. A model is similar to but simpler than the system it represents. One purpose of a model is to enable the analyst to predict the effect of changes to the system. On the one hand, a model should be a close approximation to the real system and incorporate most of its salient features. On the other hand, it should not be so complex that it is impossible to understand and experiment with it. A good model is a judicious trade-off between realism and simplicity.

□ Modelling is the process of generating abstract, conceptual, graphical and/or mathematical models. Science offers a growing collection of methods, techniques and theory about all kinds of specialized scientific modelling. Modelling also means to find relations between systems and models.

WHAT IS SIMULATION?

□ In its broadest sense, simulation is a tool to evaluate the performance of a system, existing or proposed, under different configurations of interest and over long periods of real time.

□ A simulation of a system is the operation of a model of the system. The model can be reconfigured and experimented with. Usually, this is impossible, too expensive or impractical to do in the system it represents. The operation of the model can be studied,

and hence, properties concerning the behaviour of the actual system or its subsystem can be inferred.

□ Simulation is used before an existing system is altered or a new system built, to reduce the chances of failure to meet specifications, to eliminate unforeseen bottlenecks, to prevent under or over-utilization of resources, and to optimize system performance. For instance, simulation can be used to answer questions like: What is the best design for a new telecommunications network? What are the associated resource requirements? How will a telecommunication network perform when the traffic load increases by 50%? How will a new routing algorithm affect its performance? Which network protocol optimizes network performance? What will be the impact of a link failure?

MODELING & SIMULATION

Modeling and simulation (M&S) are the use of a physical or logical representation of a given system to generate data and help determine decisions or make predictions about the system. M&S is widely used in the social and physical sciences, engineering, manufacturing and product development, among many other areas.

Applications of modeling and simulation include:

- Creating models of weather systems, simulating behaviour based on available data to generate predictive information for forecasts. A hurricane forecast model, for example, is designed to predict a given storm's track and intensity, as well as related events such as storm surges.
- Simulating the effect of severe weather events like hurricanes and storm surges on infrastructure to guide the design of more resilient systems.
- Creating a program to model a social situation and observing the behaviour of individuals in the simulation when the program runs. Social simulations can be used to yield predictive data about how things happen in real-world environments, such as how social norms develop.
- Simulating how a physical change to a system will affect its performance. For example, NASA has explored drone aerodynamics with logical models simulating air pressures and currents around the rotors. That information could potentially be used to develop designs that reduce turbulence, which would make the vehicles quieter in operation.

TYPE OF MODELS

There are many types of models and different ways of classifying/grouping them. Below are the most notable types of models. To encourage a better grasp of them, each type will be accompanied by simple examples!

Deterministic vs. Stochastic models

At the highest level, simulation models can be stochastic or deterministic. Here's how they differ:

- Deterministic models are applied when dealing with processes whose behaviour can be fully predicted from start to end. With a given set of inputs, deterministic models always produce

the same outputs. There is no randomness in deterministic models since we know the precise values of all input variables.

□ Stochastic models are applied whenever we cannot accurately estimate input variables – either because we don't have enough information about the variables or because they fluctuate within a specific range.

For Example: Suppose we have a mobile app, and we want to find out how many people will be using it in a month (using a deterministic model). If we currently have 1,000 users and know that the app will lose 20 users, attract 400 new users, and have 50 returning users within a month, we can easily calculate how many users the app will have in the future. The calculation would have this quite simple form:

Monthly Users = Current Users + New Users + Returning Users – Leaving Users

Or, in our case: $1,000 + 400 + 50 - 20 = 1,430$

Unfortunately, in reality, things are frequently much harder than that. Due to various reasons, we may not know for sure how many users we will lose and how many users we will acquire. And that's where stochastic simulation comes into play!

With stochastic simulation, we can handle uncertainties in the data through probability distributions. Once a suitable probability distribution is chosen for the target process, we can sample data from that distribution, use the data as inputs for our model, and record the model's outputs. Because we sample values for unknown variables randomly (within a specified distribution), stochastic models produce different results each time we run them.

Static vs. Dynamic models

A brief comparison of static and dynamic models is provided below:

□ Static models are applied to simulate processes at a specific instant in time. The outputs of these models only depend on the inputs and internal model variables. Static models produce a snapshot of the process in a specified timeframe. Static methods should be used to model systems that stay unchanged as time goes on, or to model systems at a specific time.

□ Dynamic models take into account not only inputs at the current time, but also outputs at previous points in time. Additionally, the output of dynamic models depends on time itself (i.e., variables in dynamic models are denoted as functions of time). Dynamic models should be used if the target process is known to change over time – perhaps due to seasonality or wear.

For Example: If assumptions were made about the sales of a seasonal product based on current market conditions and those assumptions to make predictions were attempted far into the future, then the results would be poor. Our assumptions wouldn't be able to represent the changed process next week or next month. Similarly, there's no point in applying a dynamic model to a process that doesn't change with time. Dynamic models could still yield valid results when used with static systems, but they would be overkill and require more effort than necessary to simulate a static process.

Discrete vs. Continuous models

The main distinctions between discrete and continuous models can be described as follows:

- Discrete models are applied when the specific instant at which an event occurs, with no changes occurring between events, can be pinpointed. Examples of discrete processes are app downloads, product purchases, or customer arrivals at a bank.
- Continuous models are applied to processes whose state changes continuously. Examples of continuous processes and phenomena are temperature, speed, or revenue.

In some situations, continuous and discrete models may be used interchangeably, but they may produce results that don't make sense or are not sufficiently detailed.

For Example: If we use a continuous model to simulate software purchases, the model may produce illogical outputs, like 1.5 purchases. Software purchases can be counted, making continuous models unsuitable for their representation (although we could still achieve good results with them). In contrast, if we use a discrete method to model a continuous process, we may not capture the changes occurring between particular points in time. This may lead to loss of information and thus diminish performance.

Other Types Of Models Include:

Physical Models

These are called iconic models. Good examples of physical models are model cars, model railway, model airplane, scale models, etc. A model railway can be used to study the behaviour of a real railway, also scale models can be used to study a plant layout design. In simulation studies, iconic models are rarely used.

Analogue Models

These are similar to iconic models. But here some other entities are used to represent directly the entities of the real world. An example is the analogue computer where the magnitudes of the electrical currents flowing in a circuit can be used to represent quantities of materials or people moving around in a system. Other examples are; the gauge used to check the pressure in a tyre. The movement of the dial represent the air pressure in the tyre. In medical examination, the marks of electrical current on paper, is the analogue representation of the working of muscles or organs.

Heuristic Models

These models use intuitive (or futuristic) rules with the hope that it will produce workable solutions, which can be improved upon. For example, the Arthur C Clerk's heuristic model was the forerunner of the communications satellite and today's international television broadcast.

Mathematical Models

These are models used for predictive (projecting) purposes. They are abstract and take the form of mathematical expressions of relationships. For examples:

□ $x^2 + y^2 = 1$ (mathematical model of a circle of radius 1,

□ $\text{Revenue} = \text{Price} \times \text{Quantity}$

□ $\text{Profit} = \text{Revenue} - \text{Cost}$,

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□ Linear programming models and so on.

Mathematical models can be as simple as interest earnings on a savings account or as complex as the operation of an entire factory or landing astronauts on the moon. The development of mathematical models requires great deal of skill and knowledge.

Simulation Models

Here, instead of entities being represented physically, they are represented by sequences of random numbers subject to the assumptions of the model. These models represent (emulate) the behaviour of a real system. They are used where there are no suitable mathematical models or where the mathematical model is too complex or where it is not possible to experiment upon a working system without causing serious disruption.

HOW TO DEVELOP A MODEL

The steps involved in developing a simulation model, designing a simulation experiment, and performing simulation analysis are:

Step 1. Identify the problem.

Enumerate problems with an existing system. Produce requirements for a proposed system.

Step 2. Formulate the problem.

Select the bounds of the system, the problem or a part thereof, to be studied. Define overall objective of the study and a few specific issues to be addressed. Define performance measures - quantitative criteria on the basis of which different system configurations will be compared and ranked. Identify, briefly at this stage, the configurations of interest and formulate hypotheses about system performance. Decide the time frame of the study, i.e., will the model be used for a one-time decision (e.g., capital expenditure) or over a period of time on a regular basis (e.g., air traffic scheduling). Identify the end user of the simulation model, e.g., corporate management versus a production supervisor. Problems must be formulated as precisely as possible.

Step 3. Collect and process real system data.

Collect data on system specifications (e.g., bandwidth for a communication network), input variables, as well as performance of the existing system. Identify sources of randomness in the system, i.e., the stochastic input variables. Select an appropriate input probability distribution for each stochastic input variable and estimate corresponding parameter(s).

Step 4. Formulate and develop a model.

Develop schematics and network diagrams of the system (How do entities flow through the system?). Translate these conceptual models to simulation software acceptable form. Verify that the simulation model executes as intended. Verification techniques include traces, varying input parameters over their acceptable range and checking the output, substituting constants for random variables and manually checking results, and animation.

Step 5. Validate the model.

Compare the model's performance under known conditions with the performance of the real system. Perform statistical inference tests and get the model examined by system experts. Assess the confidence that the end user places on the model and address problems if any. For major simulation studies, experienced consultants advocate a structured presentation of the model by the simulation analyst(s) before an audience of management and system experts. This

not only ensures that the model assumptions are correct, complete and consistent, but also enhances confidence in the model.

Step 6. Document model for future use.

Document objectives, assumptions and input variables in detail.

Step 7. Select appropriate experimental design.

Select a performance measure, a few input variables that are likely to influence it, and the levels of each input variable.

Step 8. Establish experimental conditions for runs.

Address the question of obtaining accurate information and the most important information from each run. Determine if the system is stationary (performance measure does not change over time) or non-stationary (performance measure changes over time). Identify output data most likely to be correlated.

Step 9. Perform simulation runs.

Perform runs according to steps 7-8 above.

Step 10. Interpret and present results.

Compute numerical estimates (e.g., mean, confidence intervals) of the desired performance measure for each configuration of interest. Test hypotheses about system performance. Construct graphical displays (e.g., pie charts, histograms) of the output data. Document results and conclusions.

Step 11. Recommend further course of action.

This may include further experiments to increase the precision and reduce the bias of estimators, to perform sensitivity analyses, etc.

BENEFITS

According to practitioners, simulation modeling and analysis, when used judiciously, makes it possible to:

- Obtain a better understanding of the system by developing a mathematical model of a system of interest, and observing the system's operation in detail over long periods of time.
- Test hypotheses about the system for feasibility.
- Compress time to observe certain phenomena over long periods or expand time to observe a complex phenomenon in detail.
- Study the effects of certain informational, organizational, environmental and policy changes on the operation of a system by altering the system's model; this can be done without disrupting the real system and significantly reduces the risk of experimenting with the real system.
- Experiment with new or unknown situations about which only weak information is available.
- Identify the "driving" variables (the ones that performance measures are most sensitive to) and the inter-relationships among them.
- Identify bottlenecks in the flow of entities (material, people, etc.) or information.

- Use multiple performance metrics for analysing system configurations.
- Employ a systems approach to problem solving.
- Develop well designed and robust systems and reduce system development time.

PITFALLS

Simulation can be a time consuming and complex exercise. The following is a checklist of pitfalls to guard against:

When the objectives are unclear.

Using simulation when an analytic solution is available and appropriate.

When a model is invalid.

When a simulation model is too complex or too simple.

When assumptions are erroneous or incorrect.

When assumptions are not documented.

When the wrong input probability distribution is used.

When the wrong performance measurements are used.

When bugs are found in the simulation program.

When there is an initial bias in the output data.

Poor communication among the personnel involved in the simulation study.